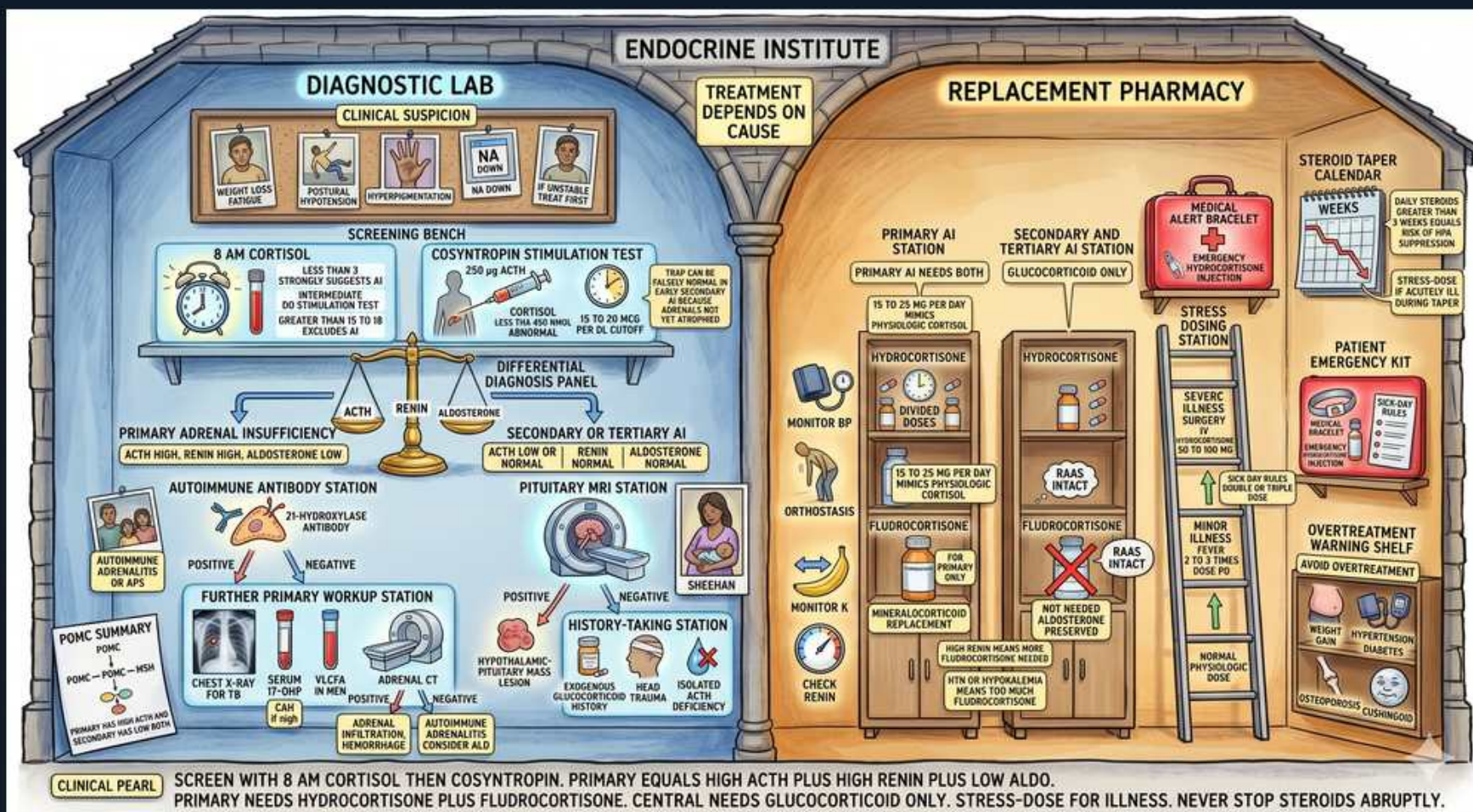


Adrenal Insufficiency — Treatment & Replacement



CLINICAL PEARL

SCREEN WITH 8 AM CORTISOL THEN COSYNTROPIN. PRIMARY EQUALS HIGH ACTH PLUS HIGH RENIN PLUS LOW ALDO. PRIMARY NEEDS HYDROCORTISONE PLUS FLUDROCORTISONE. CENTRAL NEEDS GLUCOCORTICOID ONLY. STRESS-DOSE FOR ILLNESS. NEVER STOP STEROIDS ABRUPTLY.

1. Replacement Pharmacy

WHAT YOU SEE

The entire wood-paneled right wing of the Endocrine Institute labelled REPLACEMENT PHARMACY, lined with shelves of hormone bottles — the pharmacy IS the lifelong hormone-replacement regimen.

WHAT IT ENCODES

Treatment of chronic adrenal insufficiency is lifelong physiologic hormone replacement: glucocorticoid for everyone, PLUS mineralocorticoid only when the deficiency is primary (adrenal). The goal is to mimic normal output without over-replacing, and to scale up during stress to prevent crisis.

BOARD TRAP

The replacement regimen DEPENDS on the cause — primary (Addison) and central (secondary/tertiary) AI differ in whether mineralocorticoid is needed. Getting this distinction wrong is the most tested concept in the scene.

2. Treatment depends on cause

WHAT YOU SEE

The central archway banner reading TREATMENT DEPENDS ON CAUSE, bridging the diagnostic lab and the pharmacy — it gates which wing (one drug vs two) the patient walks into.

WHAT IT ENCODES

Primary AI (Addison) = the whole cortex is destroyed → needs BOTH glucocorticoid and mineralocorticoid. Central AI (secondary/tertiary, i.e. pituitary/hypothalamic) = only ACTH is lost while the RAAS-driven zona glomerulosa keeps working → needs GLUCOCORTICOID ONLY. Mnemonic: 'Primary = 2 drugs, Central = 1 drug.'

BOARD TRAP

This is the central board distinction. Do NOT give fludrocortisone in secondary/tertiary AI — aldosterone is still being made normally.

3. Primary AI station

WHAT YOU SEE

primary AI shelf, hydrocortisone + fludrocortisone

WHAT IT ENCODES

Primary adrenal insufficiency requires hydrocortisone PLUS fludrocortisone because both the fasciculata (cortisol) and glomerulosa (aldosterone) are gone. Many patients also benefit from DHEA replacement (especially women with low libido/wellbeing).

BOARD TRAP

Primary = whole adrenal cortex destroyed → both cortisol AND aldosterone missing → renin is HIGH; titrate fludrocortisone to normalize renin, BP, and K+.

4. Hydrocortisone — glucocorticoid

WHAT YOU SEE

hydrocortisone bottle, divided doses

WHAT IT ENCODES

Hydrocortisone is the first-line glucocorticoid (~15–25 mg/day in 2–3 divided doses, largest dose on waking). It is preferred because its short half-life best mimics the cortisol circadian curve. Prednisone or dexamethasone are alternatives when once-daily dosing or adherence is needed.

BOARD TRAP

Mimic the circadian rhythm — give the LARGEST dose in the morning; late-evening dosing causes insomnia and worsens metabolic side effects.

5. Fludrocortisone — mineralocorticoid

WHAT YOU SEE

fludrocortisone bottle at primary station

WHAT IT ENCODES

Fludrocortisone (typical 0.05–0.2 mg once daily) replaces aldosterone and is needed ONLY in primary AI. Titrate against orthostatic symptoms, supine/standing BP, serum K+, and plasma renin activity (target upper-normal renin).

BOARD TRAP

Not given in central AI. Under-dosing → hyperkalemia, salt-craving, postural hypotension; over-dosing → hypertension, edema, hypokalemia.

6. Secondary/tertiary — glucocorticoid only

WHAT YOU SEE

secondary/tertiary AI shelf, glucocorticoid only, RAAS intact

WHAT IT ENCODES

Central (secondary = pituitary ACTH deficiency; tertiary = hypothalamic CRH deficiency, classically from chronic exogenous steroids) AI needs GLUCOCORTICOID ONLY. The zona glomerulosa is RAAS/K+ driven and remains intact, so aldosterone is preserved.

BOARD TRAP

Adding fludrocortisone in central AI is wrong. Also screen for OTHER pituitary hormone deficiencies (hypothyroidism, hypogonadism) — and always replace cortisol BEFORE thyroxine to avoid precipitating crisis.

7. RAAS intact (no fludro)

WHAT YOU SEE

A label reading RAAS INTACT with the fludrocortisone bottle crossed out by a red X — the intact aldosterone axis is why the mineralocorticoid is struck off in central AI.

WHAT IT ENCODES

In central AI the renin–angiotensin–aldosterone axis is preserved, so K+ stays normal and no mineralocorticoid is required. Any hyponatremia is dilutional, from cortisol deficiency removing the normal suppression of ADH.

BOARD TRAP

Hyponatremia in central AI is from cortisol deficiency (unsuppressed ADH), NOT aldosterone loss — so K+ is normal and the fix is glucocorticoid, not fludrocortisone.

8. Stress / sick-day dosing station

WHAT YOU SEE

stress dosing station, fever/illness, double-triple dose

WHAT IT ENCODES

During minor illness/fever the patient should DOUBLE or TRIPLE the oral glucocorticoid (sick-day rules) until recovered, because the failed adrenal cannot mount a stress cortisol surge. Common rule: double for fever, triple for high fever, switch to parenteral if vomiting.

BOARD TRAP

Sick-day rules apply to ALL AI types, INCLUDING secondary/central — failing to stress-dose is a leading trigger of adrenal crisis.

9. Stress dose — surgery/severe illness

WHAT YOU SEE

severe illness, IV hydrocortisone for major stress

WHAT IT ENCODES

Major surgery, trauma, or severe illness needs high-dose parenteral hydrocortisone — classically 100 mg IV bolus then 50 mg q6–8h (or 200 mg/24h continuous) — plus IV fluids, tapering as the patient recovers.

BOARD TRAP

At these high hydrocortisone doses a separate mineralocorticoid is NOT needed, because hydrocortisone itself has intrinsic mineralocorticoid activity at ≥50 mg/day.

10. Steroid taper calendar

WHAT YOU SEE

A wall calendar headed STEROID TAPER counting down over WEEKS — the slow week-by-week countdown IS the mandatory gradual taper, never an abrupt stop.

WHAT IT ENCODES

Glucocorticoids that have been given long enough to suppress the HPA axis (roughly >3 weeks of supraphysiologic dosing) must be TAPERED, never stopped abruptly, to let endogenous CRH-ACTH-cortisol recover. The suppressed axis is the cause of tertiary AI.

BOARD TRAP

Abrupt steroid withdrawal precipitates adrenal crisis — the #1 clinical pearl here, and the mechanism behind iatrogenic/tertiary adrenal insufficiency.

11. Medical alert bracelet

WHAT YOU SEE

A red MEDICAL ALERT bracelet/dog-tag icon on the shelf — the worn tag is what tells responders to give stress-dose steroids on sight.

WHAT IT ENCODES

Every AI patient should wear a medical-alert bracelet/necklace identifying glucocorticoid dependence, so that emergency responders give stress-dose steroids immediately if the patient is found obtunded or hypotensive.

BOARD TRAP

First responders must know to give parenteral stress-dose hydrocortisone empirically — treatment of suspected crisis is never delayed for confirmatory labs.

12. Patient emergency kit

WHAT YOU SEE

A red PATIENT EMERGENCY KIT case holding an injectable hydrocortisone syringe — the self-injection kit IS the crisis rescue when the patient can't take oral.

WHAT IT ENCODES

Patients are equipped with an emergency IM hydrocortisone kit (e.g. 100 mg) and trained in self-injection for crisis or when vomiting prevents oral dosing.

BOARD TRAP

Adrenal crisis management = IV/IM hydrocortisone + IV normal saline + dextrose (for hypoglycemia); give empirically BEFORE confirmatory cortisol/ACTH testing. Use dexamethasone if you must preserve a cosyntropin test, as it doesn't cross-react in the cortisol assay.

13. Avoid overtreatment

WHAT YOU SEE

The OVERTREATMENT WARNING SHELF showing the iatrogenic-Cushing face (moon facies, weight gain) — the puffy figure is the cost of over-replacing glucocorticoid.

WHAT IT ENCODES

Chronic glucocorticoid over-replacement produces iatrogenic Cushing — central weight gain, osteoporosis, hyperglycemia, hypertension, skin thinning. Use the lowest dose controlling symptoms.

BOARD TRAP

There is NO reliable lab marker for glucocorticoid dose adequacy (random cortisol/ACTH don't work on replacement) — titrate clinically to symptoms and avoid excess.

14. Never stop steroids abruptly

WHAT YOU SEE

clinical pearl banner: never stop steroids abruptly

WHAT IT ENCODES

One-line summary: primary AI = hydrocortisone + fludrocortisone; central AI = glucocorticoid only; stress-dose for illness/surgery; NEVER stop steroids abruptly.

BOARD TRAP

Sudden cessation or a missed sick-day/stress dose is the classic trigger of fatal adrenal crisis — present with hypotension/shock, hyponatremia, hyperkalemia (primary), hypoglycemia, and abdominal pain.